

class 14

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Section 1. Differential Expression Analysis

GSE37704_featurecounts.csv and GSE37704_metadata.csv. This is similar to our starting point for the last class where we used DESeq2 for the first time. We will use it again today!

```
library(DESeq2)
```

```
## Loading required package: S4Vectors
## Loading required package: stats4
## Loading required package: BiocGenerics
## Loading required package: generics
##
## Attaching package: 'generics'
## The following objects are masked from 'package:base':
##
##   as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##   setequal, union
##
## Attaching package: 'BiocGenerics'
## The following objects are masked from 'package:stats':
##
##   IQR, mad, sd, var, xtabs
## The following objects are masked from 'package:base':
##
##   anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##   colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##   get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##   order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##   rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##   unsplit, which.max, which.min
##
## Attaching package: 'S4Vectors'
## The following object is masked from 'package:utils':
##
##   findMatches
## The following objects are masked from 'package:base':
##
##   expand.grid, I, unname
```

```

## Loading required package: IRanges
##
## Attaching package: 'IRanges'
## The following object is masked from 'package:grDevices':
##
##     windows
## Loading required package: GenomicRanges
## Loading required package: Seqinfo
## Loading required package: SummarizedExperiment
## Loading required package: MatrixGenerics
## Loading required package: matrixStats
##
## Attaching package: 'MatrixGenerics'
## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars
## Loading required package: Biobase
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase)", and for packages 'citation("pkgname)".
##
## Attaching package: 'Biobase'
## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians
## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians
counts <- read.csv("GSE37704_featurecounts.csv")
metadata <- read.csv("GSE37704_metadata.csv")

```

Check on data structure

```
head(counts)
```

```
##           ensgene length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
## 1 ENSG00000186092    918          0          0          0          0          0
## 2 ENSG00000279928    718          0          0          0          0          0
## 3 ENSG00000279457   1982         23         28         29         29         28
## 4 ENSG00000278566    939          0          0          0          0          0
## 5 ENSG00000273547    939          0          0          0          0          0
## 6 ENSG00000187634   3214        124        123        205        207        212
##   SRR493371
## 1          0
## 2          0
## 3         46
## 4          0
## 5          0
## 6        258
```

```
metadata
```

```
##           id      condition
## 1 SRR493366 control_sirna
## 2 SRR493367 control_sirna
## 3 SRR493368 control_sirna
## 4 SRR493369   hoxa1_kd
## 5 SRR493370   hoxa1_kd
## 6 SRR493371   hoxa1_kd
```

```
ncol(counts)
```

```
## [1] 8
```

```
nrow(metadata)
```

```
## [1] 6
```

```
cleancounts <- counts[, -c(1, 2)]
```

```
colnames(cleancounts)
```

```
## [1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
metadata$id
```

```
## [1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

Q. Complete the code below to remove the troublesome first column from countData

```
all(colnames(cleancounts) == metadata$id)
```

```
## [1] TRUE
```

```
head(cleancounts)
```

```
##   SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
## 1          0          0          0          0          0          0
## 2          0          0          0          0          0          0
## 3         23         28         29         29         28         46
## 4          0          0          0          0          0          0
## 5          0          0          0          0          0          0
## 6        124        123        205        207        212        258
```

Remove zero count genes

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```
to.keep.inds <- rowSums(cleancounts) > 0
```

```
nonzero_counts <- cleancounts[to.keep.inds, ]
```

```
head(nonzero_counts)
```

```
##      SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
## 3           23          28          29          29          28          46
## 6           124         123         205         207         212         258
## 7           1637        1831        2383        1226        1326        1504
## 8           120         153         180         236         255         357
## 9            24          48          65          44          48          64
## 10           4           9          16          14          16          16
```

Running DESeq2

Set up DESeq object

```
dds <- DESeqDataSetFromMatrix(
  countData = nonzero_counts,
  colData    = metadata,
  design     = ~ condition
)
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

Run DESeq

```
dds <- DESeq(dds)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

Get results

```
res <- results(dds)
```

```
head(res)
```

```
## log2 fold change (MLE): condition hoxa1 kd vs control sirna
```

```
## Wald test p-value: condition hoxa1 kd vs control sirna
```

```
## DataFrame with 6 rows and 6 columns
```

```
##      baseMean log2FoldChange      lfcSE      stat      pvalue      padj
##      <numeric>      <numeric> <numeric> <numeric>      <numeric>      <numeric>
## 3      29.9136      0.1792571 0.3248216    0.551863 5.81042e-01 6.86555e-01
## 6      183.2296      0.4264571 0.1402658    3.040350 2.36304e-03 5.15718e-03
## 7     1651.1881     -0.6927205 0.0548465   -12.630158 1.43990e-36 1.76549e-35
## 8      209.6379      0.7297556 0.1318599     5.534326 3.12428e-08 1.13413e-07
## 9       47.2551      0.0405765 0.2718928     0.149237 8.81366e-01 9.19031e-01
```

```
## 10    11.9798      0.5428105 0.5215598    1.040744 2.97994e-01 4.03379e-01
```

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

```
##
## out of 15975 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 4349, 27%
## LFC < 0 (down)    : 4396, 28%
## outliers [1]      : 0, 0%
## low counts [2]    : 1237, 7.7%
## (mean count < 0)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

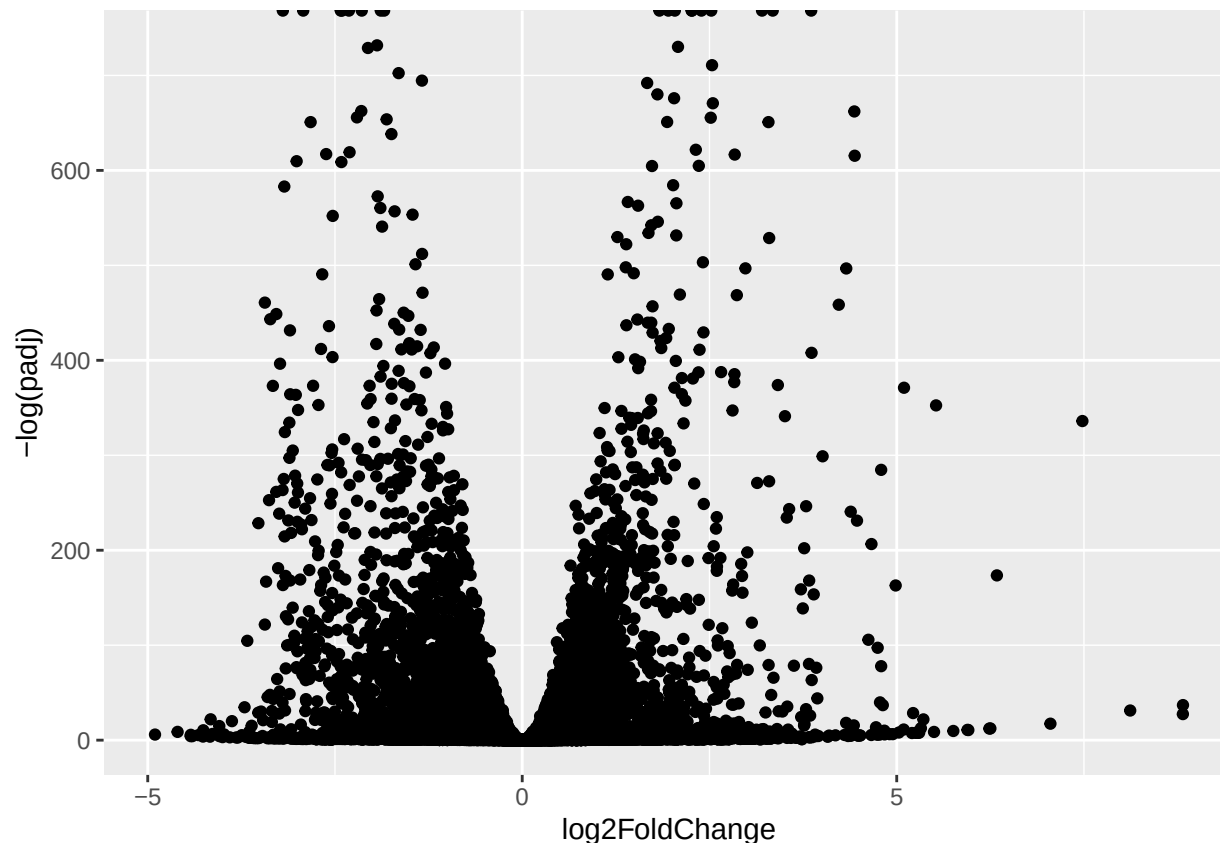
4349 genes upregulated and 4396 genes down regulated.

Volcano plot

```
library(ggplot2)
```

```
ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point()
```

```
## Warning: Removed 1237 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
# Convert DESeq2 results to data frame for ggplot
res_df <- as.data.frame(res)

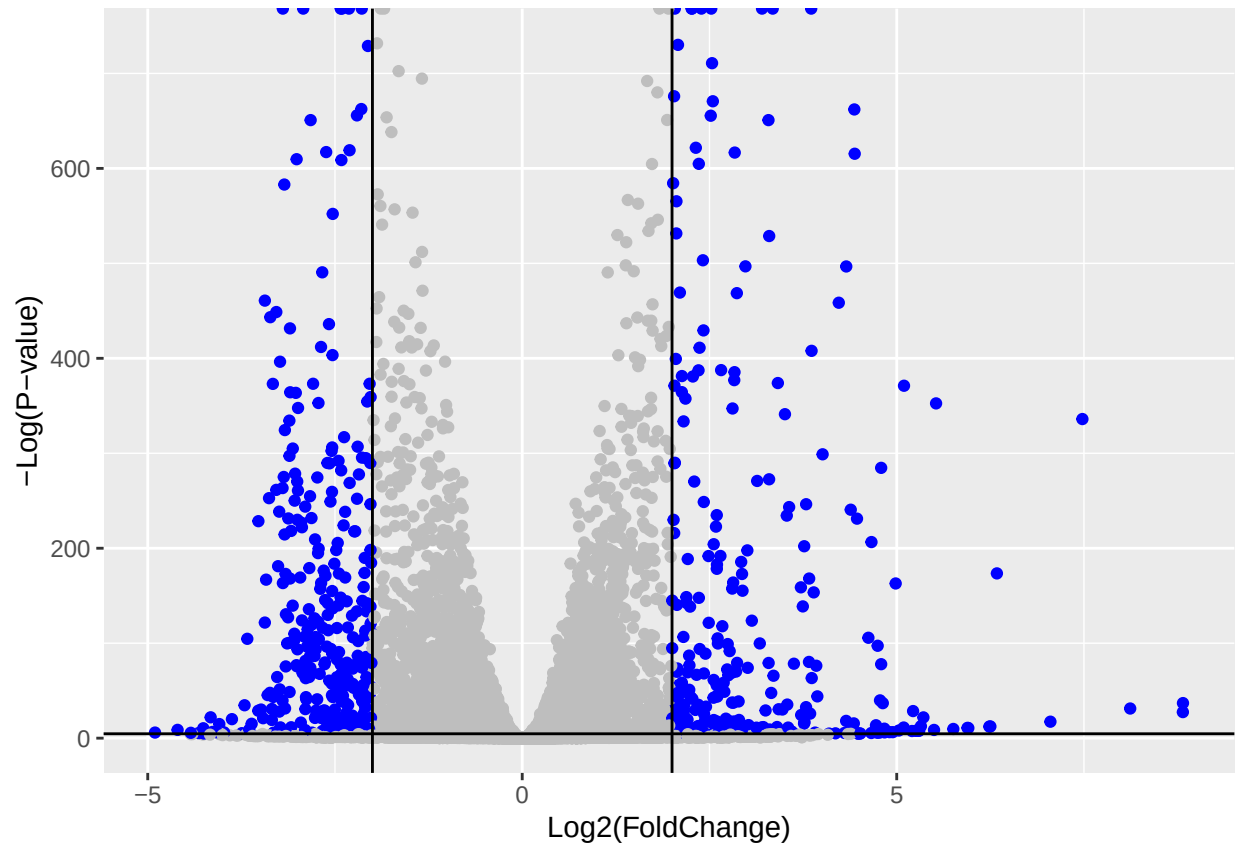
# Make a color vector for all genes
mycols <- rep("gray", nrow(res_df))

# Color blue the genes with fold change above 2
mycols[ abs(res_df$log2FoldChange) > 2 ] <- "blue"

# Color gray those with adjusted p-value more than 0.01
mycols[ res_df$padj > 0.01 ] <- "gray"

ggplot(res_df) +
  aes(x = log2FoldChange,
      y = -log(padj)) +
  geom_point(color = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log(0.01))
```

```
## Warning: Removed 1237 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Add gene annotation

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
##
```

```
library("AnnotationDbi")
library("org.Hs.eg.db")

head(
  mapIds(x = org.Hs.eg.db,
        keys = row.names(res),
        column = "SYMBOL",
        keytype = "ENTREZID",
        multiVals = "first"),
  10
)
```

```
## 'select()' returned 1:1 mapping between keys and columns
```

```
##      3      6      7      8      9      10      11
##      NA      NA      NA      NA    "NAT1"    "NAT2"    "NATP"
##     12     13     14
## "SERPINA3"  "AADAC"  "AAMP"
```

```
res$symbol <- mapIds(
  x = org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENTREZID",
  column = "SYMBOL",
  multiVals = "first"
)
```

'select()' returned 1:1 mapping between keys and columns

```
res$entrez <- mapIds(
  x = org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENTREZID",
  column = "ENTREZID",
  multiVals = "first"
)
```

```
res$name <- mapIds(org.Hs.eg.db,
  keys=row.names(res),
  keytype="ENTREZID",
  column="GENENAME",
  multiVals="first")
```

'select()' returned 1:1 mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

##	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
##	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
## 3	29.913579	0.1792571	0.3248216	0.551863	5.81042e-01	6.86555e-01
## 6	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03	5.15718e-03
## 7	1651.188076	-0.6927205	0.0548465	-12.630158	1.43990e-36	1.76549e-35
## 8	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08	1.13413e-07
## 9	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01	9.19031e-01
## 10	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01	4.03379e-01
## 11	108.922128	2.0570638	0.1969053	10.446970	1.51282e-25	1.30538e-24
## 12	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02	2.37452e-02
## 13	9128.439422	0.3899088	0.0467163	8.346304	7.04321e-17	4.21963e-16
## 14	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01	NA
##	symbol	entrez	name			
##	<character>	<character>	<character>			
## 3	NA	3	NA			
## 6	NA	6	NA			
## 7	NA	7	NA			
## 8	NA	8	NA			
## 9	NAT1	9	N-acetyltransferase 1			
## 10	NAT2	10	N-acetyltransferase 2			
## 11	NATP	11	N-acetyltransferase ..			
## 12	SERPINA3	12	serpin family A memb..			
## 13	AADAC	13	arylacetamide deacet..			
## 14	AAMP	14	angio associated mig..			

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$pvalue),]  
write.csv(res, file="deseq_results.csv")
```

Section 2. Pathway Analysis

run gage analysis

```
library(pathview)
```

```
## #####  
## Pathview is an open source software package distributed under GNU General  
## Public License version 3 (GPLv3). Details of GPLv3 is available at  
## http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to  
## formally cite the original Pathview paper (not just mention it) in publications  
## or products. For details, do citation("pathview") within R.  
##  
## The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG  
## license agreement (details at http://www.kegg.jp/kegg/legal.html).  
## #####
```

```
library(gage)
```

```
##
```

```
library(gageData)
```

```
data(kegg.sets.hs)  
data(sigmet.idx.hs)  
# Focus on signaling and metabolic pathways only  
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]  
# Examine the first 3 pathways  
head(kegg.sets.hs, 3)
```

```
## $`hsa00232 Caffeine metabolism`  
## [1] "10" "1544" "1548" "1549" "1553" "7498" "9"  
##  
## $`hsa00983 Drug metabolism - other enzymes`  
## [1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"  
## [9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"  
## [17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"  
## [25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"  
## [33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"  
## [41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"  
## [49] "8824" "8833" "9" "978"  
##  
## $`hsa00230 Purine metabolism`  
## [1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"  
## [9] "108" "10846" "109" "111" "11128" "11164" "112" "113"  
## [17] "114" "115" "122481" "122622" "124583" "132" "158" "159"  
## [25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"  
## [33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"  
## [41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"  
## [49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"  
## [57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
```

```
## [65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
## [73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
## [81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
## [89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
## [97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
## [105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
## [113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
## [121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
## [129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
## [137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
## [145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
## [153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
## [161] "9583" "9615"
```

We need a named vector as input for gage

```
Foldchanges = res$log2FoldChange
names(Foldchanges) = res$entrez
head(Foldchanges)
```

```
##      898      1052      1687      2283      5358      6064
## -2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

Now, let's run the gage pathway analysis.

```
# Get the results
keggres = gage(Foldchanges, gsets=kegg.sets.hs)
```

```
attributes(keggres)
```

```
## $names
## [1] "greater" "less" "stats"
```

```
#Look at the first few down (less) pathways
head(keggres$less)
```

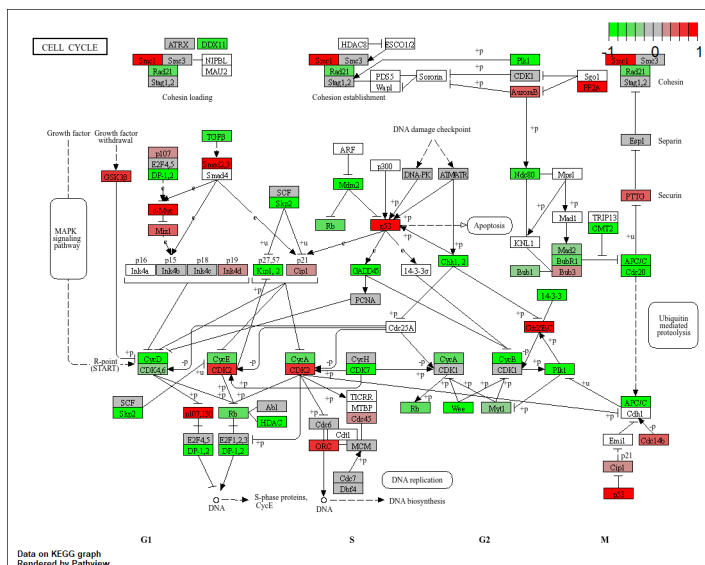
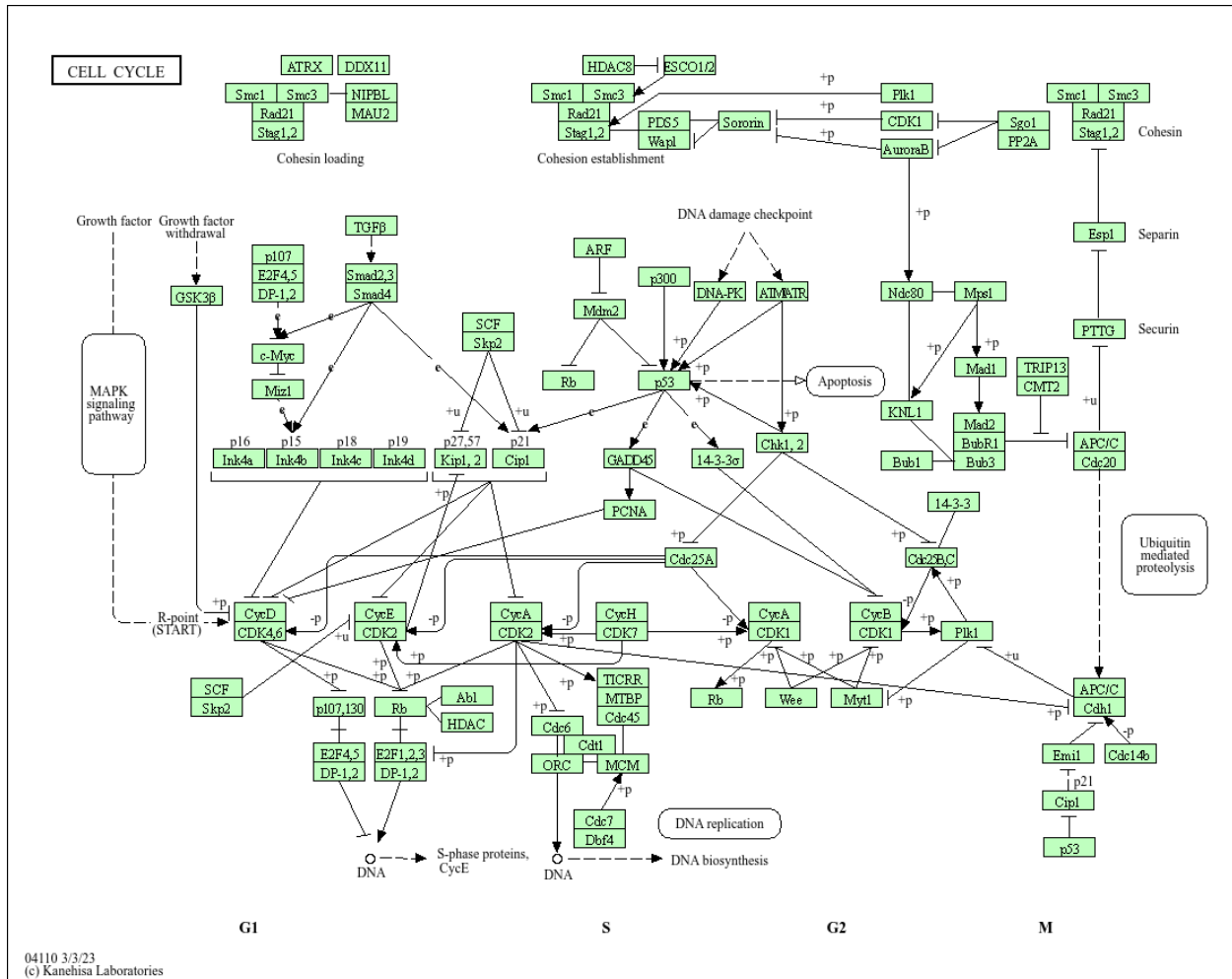
```
##                p.geomean stat.mean      p.val
## hsa03030 DNA replication      0.01885846 -2.173571 0.01885846
## hsa03430 Mismatch repair      0.04579401 -1.743605 0.04579401
## hsa03410 Base excision repair  0.05908254 -1.598886 0.05908254
## hsa04120 Ubiquitin mediated proteolysis 0.06440557 -1.528736 0.06440557
## hsa00360 Phenylalanine metabolism 0.11126857 -1.250743 0.11126857
## hsa04510 Focal adhesion      0.12267443 -1.164165 0.12267443
##                q.val set.size      exp1
## hsa03030 DNA replication      0.8745475      17 0.01885846
## hsa03430 Mismatch repair      0.8745475      16 0.04579401
## hsa03410 Base excision repair  0.8745475      20 0.05908254
## hsa04120 Ubiquitin mediated proteolysis 0.8745475      65 0.06440557
## hsa00360 Phenylalanine metabolism 0.8745475      14 0.11126857
## hsa04510 Focal adhesion      0.8745475     142 0.12267443
```

```
pathview(gene.data=Foldchanges, pathway.id="hsa04110")
```

```
## 'select()' returned 1:1 mapping between keys and columns
```

```
## Info: Working in directory C:/Users/chenr/OneDrive/Desktop/Class 14
```

```
## Info: Writing image file hsa04110.pathview.png
```



A different PDF based output of the same data
 pathview(gene.data=Foldchanges, pathway.id="hsa04110", kegg.native=FALSE)

'select()' returned 1:1 mapping between keys and columns

```
## Warning: reconcile groups sharing member nodes!
##      [,1] [,2]
## [1,] "9"  "300"
## [2,] "9"  "306"

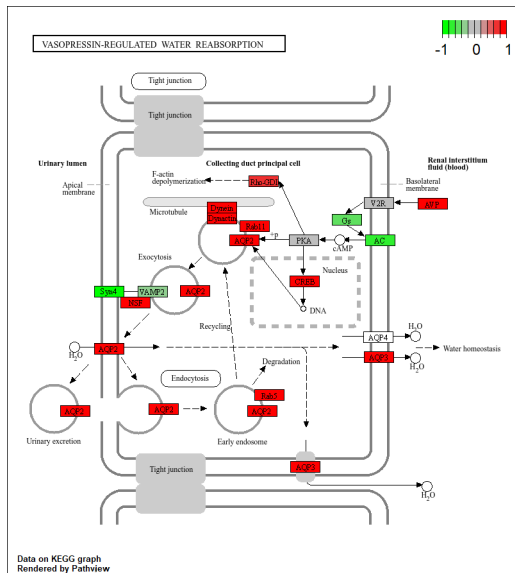
## Info: Working in directory C:/Users/chenr/OneDrive/Desktop/Class 14
## Info: Writing image file hsa04110.pathview.pdf
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]
head(keggrespathways)

## [1] "hsa04962 Vasopressin-regulated water reabsorption"
## [2] "hsa02010 ABC transporters"
## [3] "hsa04360 Axon guidance"
## [4] "hsa04974 Protein digestion and absorption"
## [5] "hsa00970 Aminoacyl-tRNA biosynthesis"
## Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids

## [1] "hsa04962" "hsa02010" "hsa04360" "hsa04974" "hsa00970"

Finally, lets pass these IDs in keggresids to the pathview() function to draw plots for all the top 5 pathways.
pathview(gene.data=Foldchanges, pathway.id=keggresids, species="hsa")

## 'select()' returned 1:1 mapping between keys and columns
## Info: Working in directory C:/Users/chenr/OneDrive/Desktop/Class 14
## Info: Writing image file hsa04962.pathview.png
## 'select()' returned 1:1 mapping between keys and columns
## Info: Working in directory C:/Users/chenr/OneDrive/Desktop/Class 14
## Info: Writing image file hsa02010.pathview.png
## 'select()' returned 1:1 mapping between keys and columns
## Info: Working in directory C:/Users/chenr/OneDrive/Desktop/Class 14
## Info: Writing image file hsa04360.pathview.png
## 'select()' returned 1:1 mapping between keys and columns
## Info: Working in directory C:/Users/chenr/OneDrive/Desktop/Class 14
## Info: Writing image file hsa04974.pathview.png
## 'select()' returned 1:1 mapping between keys and columns
## Info: Working in directory C:/Users/chenr/OneDrive/Desktop/Class 14
## Info: Writing image file hsa00970.pathview.png
```



ABC TRANSPORTERS

Prokaryotic-type ABC transporters

Mineral and organic ion transporters

Sulfate / Thiosulfate	CysP CysW	CysU CysW	CysA
Tungstate	TunA	TunB	TunC
Molybdate / Tungstate	WolA	WolB	WolC
Nitrate / Nitrite / Cyanate	NitA	NitB	NitC
Bicarbonate	CysA	CysB	CysC
Taurine	TauA	TauB	TauC
Alkanesulfonate	SesA	SesB	SesC
HMP / FAMP	ThiA	ThiB	ThiC
Phthalate	CysA	CysB	CysC
Molybdate	ModA	ModB	ModC
Iron (III)	AtrA	AtrB	AtrC
Thiamine	ThiA	ThiB	ThiC
Spermidine / Putrescine	PutA	PutB	PutC
Putrescine	PutA	PutB	PutC
Manitopine	ManA	ManB	ManC
2-Aminothiophosphonate	PhaA	PhaB	PhaC
Glycine betaine / Proline	ProA	ProB	ProC
Osmoprotectant	OsmA	OsmB	OsmC

Oligosaccharide, polyol, and lipid transporters

Maltose / Maltodextrin	MaltA	MaltB	MaltC
Galactose oligomer	GalA	GalB	GalC
Maltotriose	MaltA	MaltB	MaltC
Raffinose / Stachyose	RafA	RafB	RafC
Lactose	LacA	LacB	LacC
Sorbitol / Mannitol	SorA	SorB	SorC
α -Glucoside	AgA	AgB	AgC
Oligosaccharin	OligA	OligB	OligC
α -1,4-Digalacturonate	GalA	GalB	GalC
Aldaric acid	LacA	LacB	LacC
Trehalose / Maltose	TreA	TreB	TreC
Trehalose	TreA	TreB	TreC
N-Acetylglucosamine	NagA	NagB	NagC
Cellobiose	CelA	CelB	CelC
Chitin	ChiA	ChiB	ChiC
Arabinoside	ArbA	ArbB	ArbC
Xylobiose	XylA	XylB	XylC
Sugar	SugA	SugB	SugC
Multiple sugar	MulA	MulB	MulC
Phospholipid	PhoA	PhoB	PhoC
Nucleoside	NucA	NucB	NucC
Glucose / Arabinose	GlxA	GlxB	GlxC
Glucose / Mannose	ManA	ManB	ManC
Ribose / Arabinose 2 / D-Xylose	RibA	RibB	RibC
L-Arabinose	LarA	LarB	LarC
Galactofuranose	GalA	GalB	GalC
Methyl-galactoside	MgaA	MgaB	MgaC
D-Xylose	XylA	XylB	XylC
D-Allose	AlaA	AlaB	AlaC
Fructose	FruA	FruB	FruC
Rhamnose	RhaA	RhaB	RhaC
Erythritol	EryA	EryB	EryC
Xylitol	XylA	XylB	XylC
myo-Inositol	InsA	InsB	InsC
myo-Inositol 1-phosphate	InsA	InsB	InsC
Glycerol	GlyA	GlyB	GlyC
m-Glycerol 3-phosphate	GlyA	GlyB	GlyC

Phosphate and amino acid transporters

Phosphate	PhoA	PhoB	PhoC
Phosphonate	PhoA	PhoB	PhoC
Lysine / Arginine / Ornithine	LysA	LysB	LysC
Histidine	HisA	HisB	HisC
Glutamine	GlnA	GlnB	GlnC
Aspartate / Glutamate	AspA	AspB	AspC
Arginine	ArgA	ArgB	ArgC
Glutamate / Aspartate	GluA	GluB	GluC
Octopine / Nopaline	OctA	OctB	OctC
General L-Amino acid	LysA	LysB	LysC
Glutamate	GlnA	GlnB	GlnC
Cysteine	CysA	CysB	CysC
Cysteine	CysA	CysB	CysC
S-Methylcysteine	SMcA	SMcB	SMcC
Arginine / Ornithine	ArgA	ArgB	ArgC
Arginine / Lysine / Histidine / Glutamine	ArgA	ArgB	ArgC
Arginine / Lysine / Histidine	ArgA	ArgB	ArgC
Lysine	LysA	LysB	LysC
Lysine / Arginine / Ornithine / Histidine / Octopine	LysA	LysB	LysC
Hydroxyproline	HydA	HydB	HydC
Branched-chain amino acid	LeuA	LeuB	LeuC
Neutral amino acid / Histidine	HisA	HisB	HisC
D-Methionine	MetA	MetB	MetC
Urea	UreA	UreB	UreC

ABC-2-type components without transporting function

Bactracin	BacA	BacB	BacC
Cationic antimicrobial peptide (CAMP)	CampA	CampB	CampC
Cu ²⁺ -processing for NED reductase	CuA	CuB	CuC
Cellar division involvement	YnfA	YnfB	YnfC
Actin utilization	YnfA	YnfB	YnfC

Peptide and nickel transporters

Oligopeptide	OligA	OligB	OligC
Dipeptide / Histidine / 6-Aminocaproic acid	DipA	DipB	DipC
Dipeptide	DipA	DipB	DipC
Defensin	DefA	DefB	DefC
Nickel	NicA	NicB	NicC
Glutathione	GluA	GluB	GluC
Microcin C	YnfA	YnfB	YnfC

Metallic cation, iron-sulfur and vitamin B12 transporters

Fe(II) dicarboxylate	FeA	FeB	FeC
Fe(II) hydroxamate	PhoA	PhoB	PhoC
Vitamin B12	B12A	B12B	B12C
Manganese	MnA	MnB	MnC
Manganese	MnA	MnB	MnC
Zinc	ZnA	ZnB	ZnC
Iron (II, III) / Copper / Manganese / Zinc	MnA	MnB	MnC
Iron (II) / Manganese	MnA	MnB	MnC
Manganese / Zinc	MnA	MnB	MnC
Zinc / Manganese / Iron (II)	MnA	MnB	MnC
Cobalt	CobA	CobB	CobC
Nickel	NicA	NicB	NicC
Biotin	BioA	BioB	BioC
Biotin	BioA	BioB	BioC
Antisense 2	AntA	AntB	AntC
Riboflavin	RibA	RibB	RibC

Eukaryotic-type ABC transporters

ABC A Subfamily

ABC A1	ABC A1
ABC A2	ABC A2
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ABC A100	ABC A100

ABC B Subfamily

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ABC C Subfamily

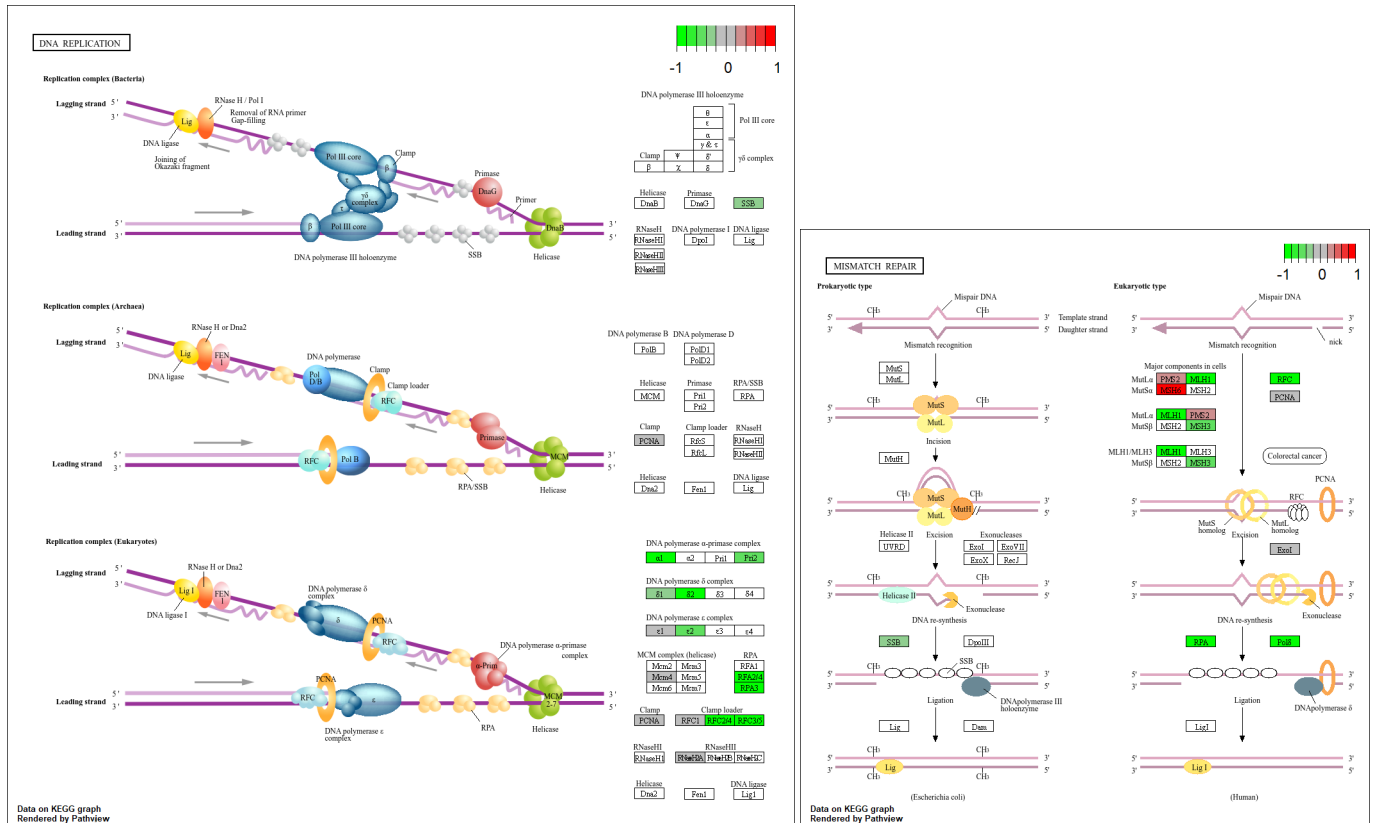
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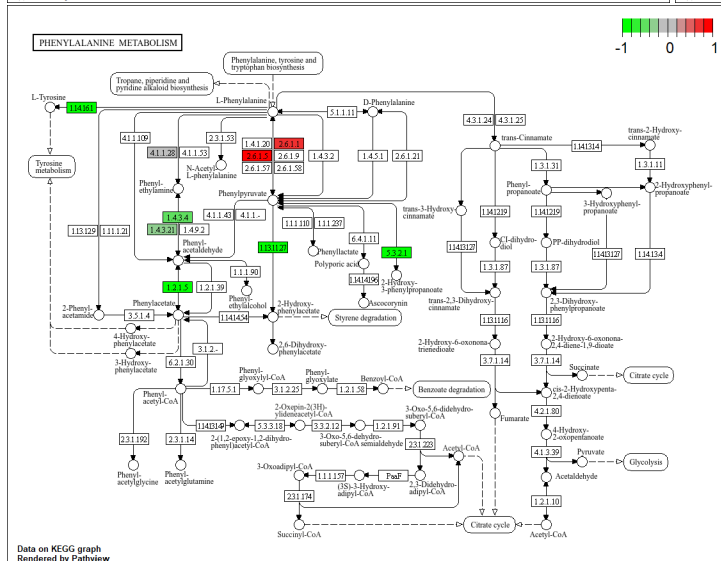
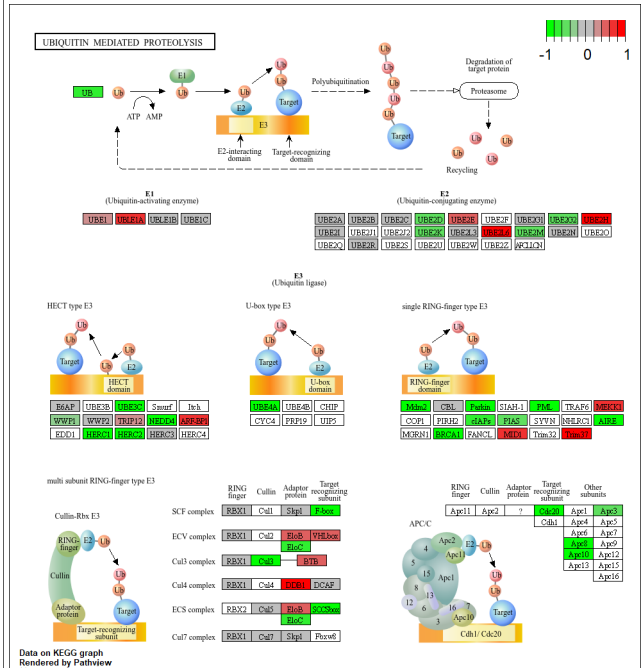
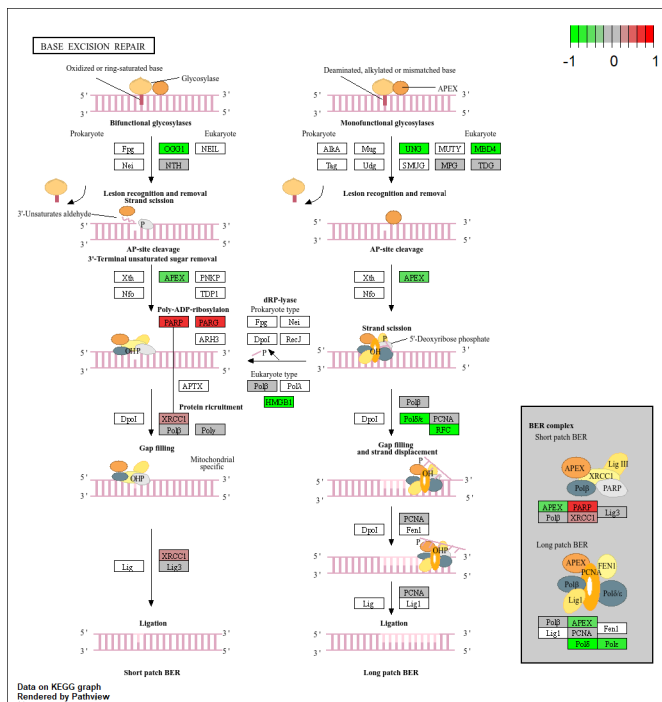
ABC D Sub


```

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## Info: Writing image file hsa00360.pathview.png

```





```

##                                     p.geomean stat.mean
## G0:0007586 digestion                0.007434165  2.462828
## G0:0044243 multicellular organismal catabolic process 0.009972396  2.437602
## G0:0007620 copulation                0.012658279  2.339581
## G0:0000187 activation of MAPK activity 0.015463270  2.176715
## G0:0006275 regulation of DNA replication 0.024352186  1.994989
## G0:0030574 collagen catabolic process 0.025208044  2.038387
##                                     p.val      q.val
## G0:0007586 digestion                0.007434165  0.9434944
## G0:0044243 multicellular organismal catabolic process 0.009972396  0.9434944
## G0:0007620 copulation                0.012658279  0.9434944
## G0:0000187 activation of MAPK activity 0.015463270  0.9434944
## G0:0006275 regulation of DNA replication 0.024352186  0.9434944
## G0:0030574 collagen catabolic process 0.025208044  0.9434944
##                                     set.size      exp1
## G0:0007586 digestion                80 0.007434165
## G0:0044243 multicellular organismal catabolic process 20 0.009972396
## G0:0007620 copulation                18 0.012658279
## G0:0000187 activation of MAPK activity 84 0.015463270
## G0:0006275 regulation of DNA replication 54 0.024352186
## G0:0030574 collagen catabolic process 16 0.025208044
##
## $less
##                                     p.geomean stat.mean
## G0:0030823 regulation of cGMP metabolic process 0.002716208 -3.086595
## G0:0030826 regulation of cGMP biosynthetic process 0.006476850 -2.728670
## G0:0006182 cGMP biosynthetic process 0.009829755 -2.448640
## G0:0046068 cGMP metabolic process 0.010028138 -2.409060
## G0:0034329 cell junction assembly 0.013213017 -2.235413
## G0:0018205 peptidyl-lysine modification 0.014979994 -2.192053
##                                     p.val      q.val
## G0:0030823 regulation of cGMP metabolic process 0.002716208 0.8422634
## G0:0030826 regulation of cGMP biosynthetic process 0.006476850 0.8422634
## G0:0006182 cGMP biosynthetic process 0.009829755 0.8422634
## G0:0046068 cGMP metabolic process 0.010028138 0.8422634
## G0:0034329 cell junction assembly 0.013213017 0.8422634
## G0:0018205 peptidyl-lysine modification 0.014979994 0.8422634
##                                     set.size      exp1
## G0:0030823 regulation of cGMP metabolic process 12 0.002716208
## G0:0030826 regulation of cGMP biosynthetic process 11 0.006476850
## G0:0006182 cGMP biosynthetic process 18 0.009829755
## G0:0046068 cGMP metabolic process 24 0.010028138
## G0:0034329 cell junction assembly 108 0.013213017
## G0:0018205 peptidyl-lysine modification 74 0.014979994
##
## $stats
##                                     stat.mean      exp1
## G0:0007586 digestion                2.462828  2.462828
## G0:0044243 multicellular organismal catabolic process 2.437602  2.437602
## G0:0007620 copulation                2.339581  2.339581
## G0:0000187 activation of MAPK activity 2.176715  2.176715
## G0:0006275 regulation of DNA replication 1.994989  1.994989
## G0:0030574 collagen catabolic process 2.038387  2.038387

```

Reactome Analysis

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))

## [1] "Total number of significant genes: 8147"

write.table(sig_genes, file="significant_genes.txt",
            row.names = FALSE, col.names=FALSE, quote=FALSE)
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The most significant pathway is Cytokine Signaling in Immune system.

It does not fully match my KEGG results, because KEGG showed pathways like DNA replication and mismatch repair. This difference can happen because Reactome and KEGG group genes into pathways differently. They also use slightly different analysis methods and pathway databases.